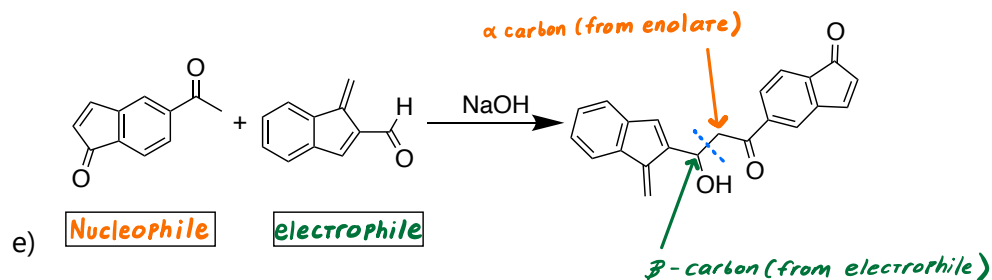
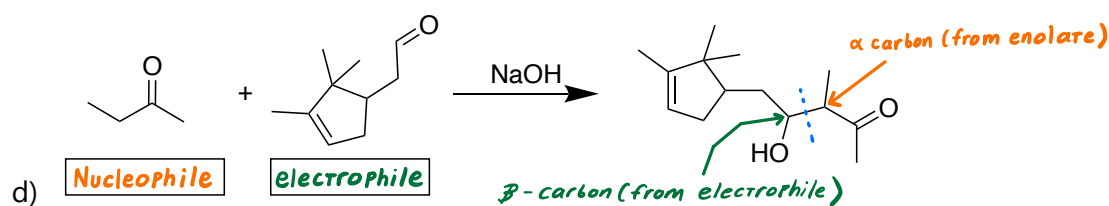
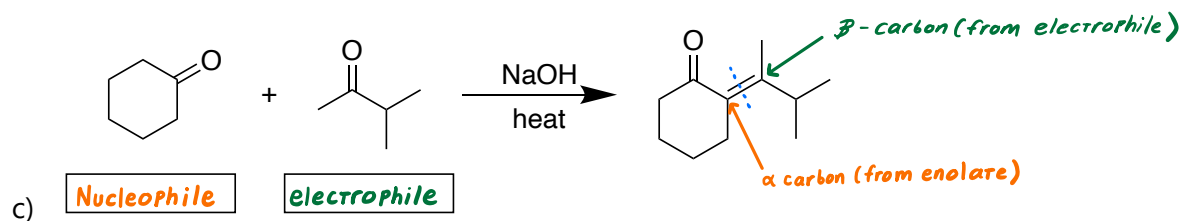
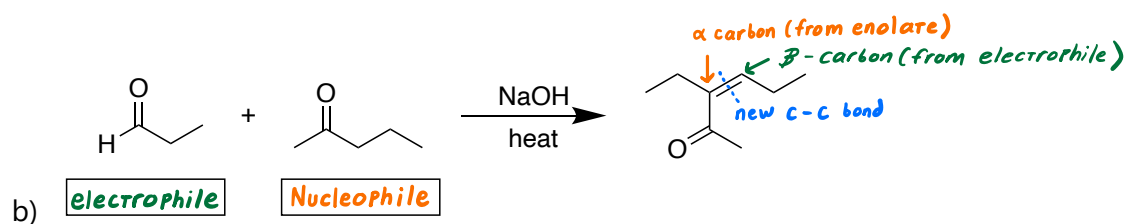
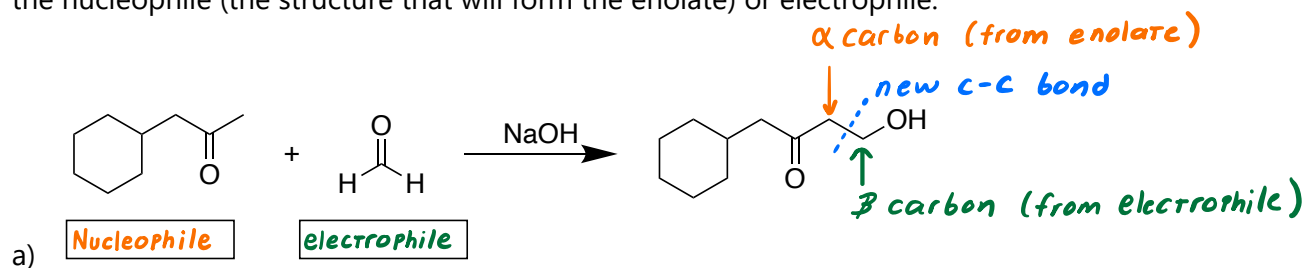


Chem 0345 – Organic Lab

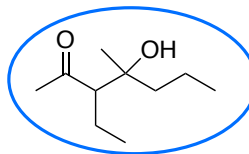
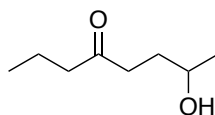
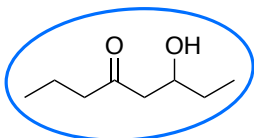
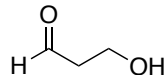
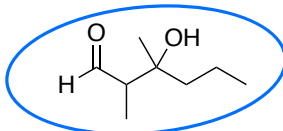
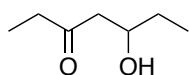
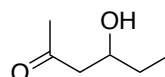
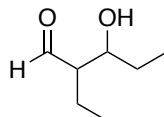
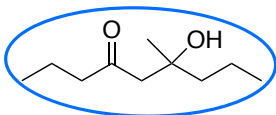
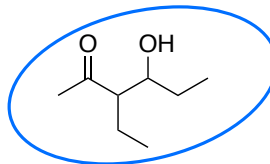
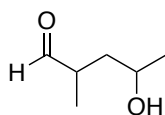
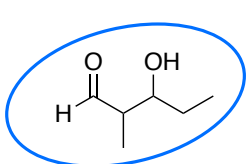
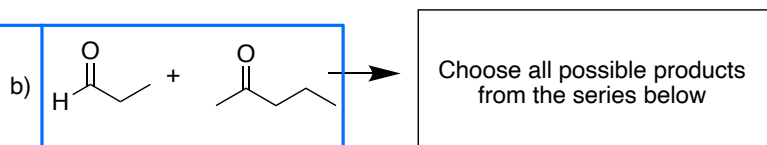
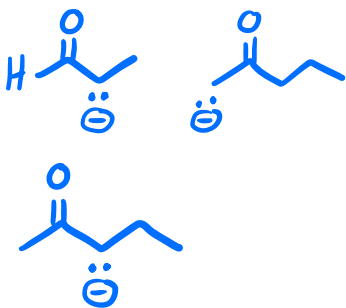
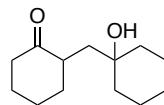
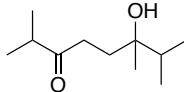
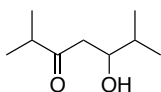
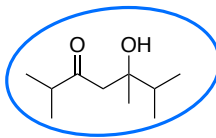
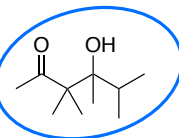
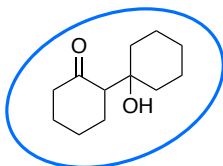
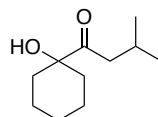
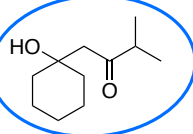
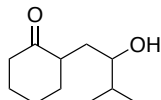
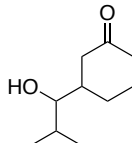
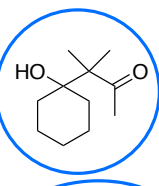
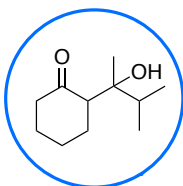
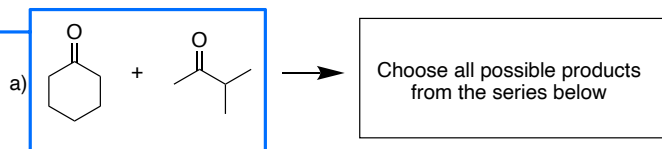
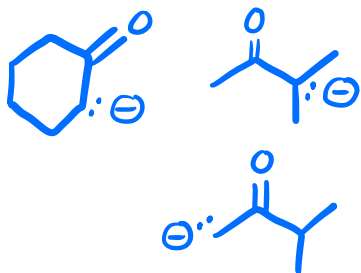
Activity 10: Aldol Addition and Aldol Condensation

Extra Practice Problems

1. Based on the structure of the product in each of the following reactions, label each reactant as the nucleophile (the structure that will form the enolate) or electrophile.



2. Select **all possible** products that could be formed in each of the following reactions.



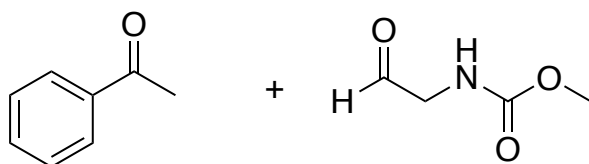
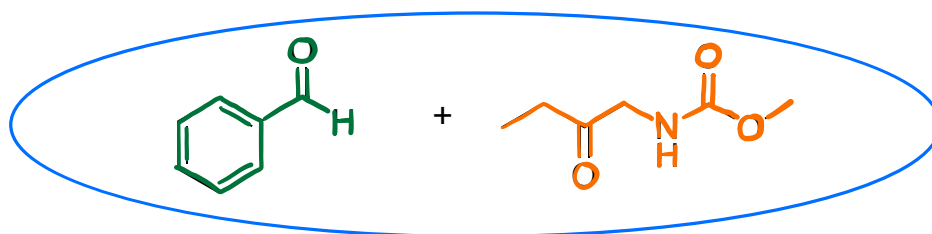
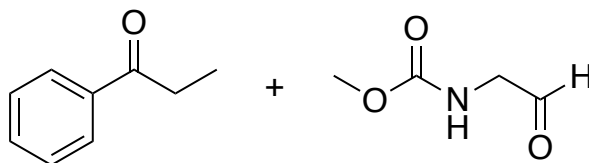
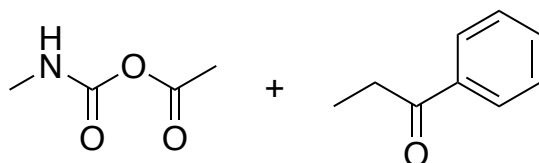
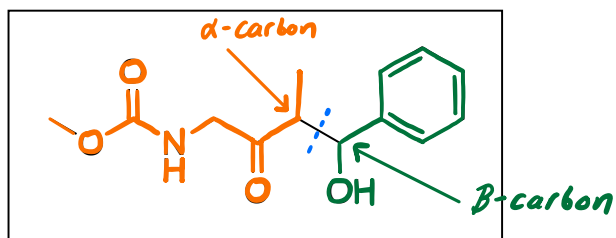
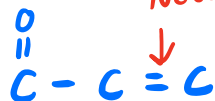
3. Select a reaction that can be used to synthesize the product shown in the box.

Look for:

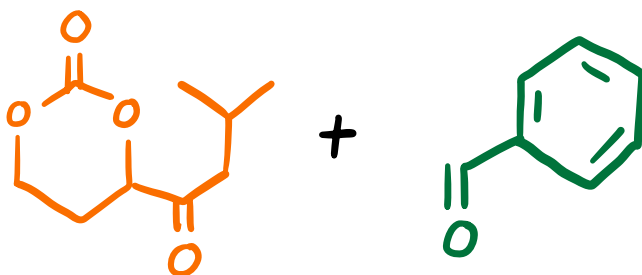
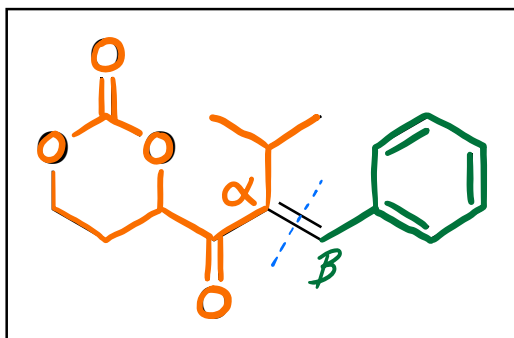


or

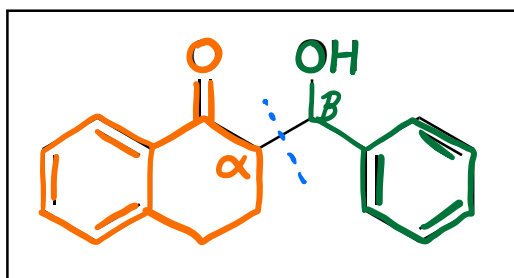
New C-C bond



4. Draw the reactants that can be used to synthesize the product shown in the box.

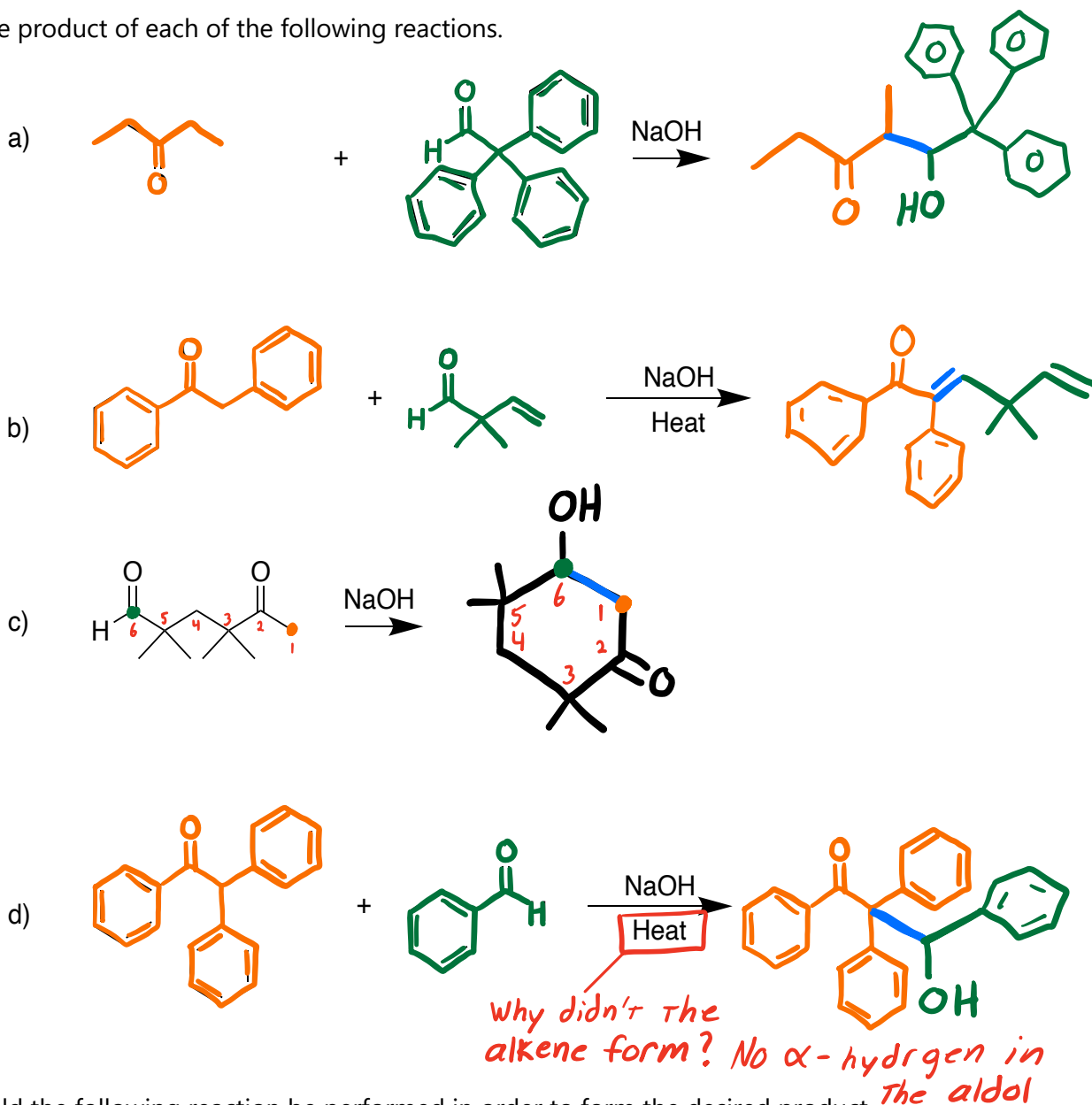


5. Draw the reactants that can be used to synthesize the product shown in the box.

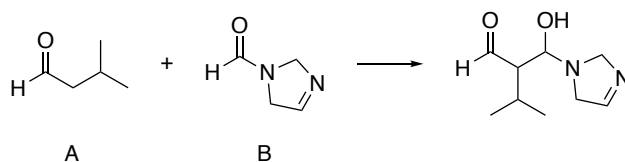


6. Predict the product of each of the following reactions.

★ Blue indicates the new C-C bond in each

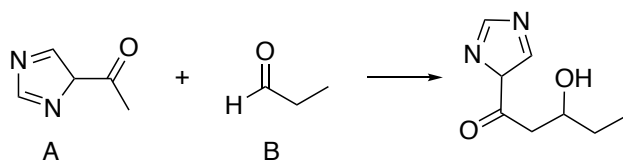


7. How should the following reaction be performed in order to form the desired product shown?



- Mix A and B in a flask with $\text{NaOH}_{(\text{aq})}$
- 1) Mix A in a flask with $\text{NaOH}_{(\text{aq})}$; 2) Slowly add B
- 1) Mix B in a flask with $\text{NaOH}_{(\text{aq})}$; 2) Slowly add A
- It is not possible to generate the product shown with reactants A and B

8. How should the following reaction be performed in order to form the desired product shown?



- Mix A and B in a flask with $\text{NaOH}_{(\text{aq})}$
- 1) Mix A and B in a flask with LDA; 2) Quench with acid
- 1) Slowly add B to a flask of LDA; 2) Slowly add in A; 3) Quench with acid
- 1) Slowly add A to a flask of LDA; 2) Slowly add in B; 3) Quench with acid